

RESEARCH ARTICLE

Real-World Parkinson's Hand Tremor Detection Using Ensemble Learning Techniques

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ABSTRACT Hand tremor is a major motor symptom of Parkinson's disease (PD) that seriously affects the life quality of patients. Conventional tremor detection methods have mainly focused on resting state. However, resting tremor is not always sufficient to accurately distinguish PD tremor from other tremor disorders such as essential tremor. Examining tremor during movement provides additional diagnostic value, but also has challenges. In this paper, we propose a novel approach to detect Parkinsonian hand tremor at walking state by exploiting the periodic synchronization between arm swings and leg steps that is observed in natural walking, specifically by using a wrist-worn sensor to capture the characteristic hand tremor that is induced during walking in Parkinson's patients. Our method first applies a dynamic scanning mechanism to screen out valid walking fragments from whole walking sequence. We then exploit walking periodicity to combine time and frequency domain features, and design two new accelerometer-based features, Average Peak Interval Time (APIT) and Average Peak Frequency (APF), which capture the characteristic amplitude and frequency patterns of PD tremors during natural walking movements. Finally, unlike traditional approaches that use uniform feature weighting, our proposed variance adaptive AdaBoost (VA-AdaBoost) algorithm uniquely incorporates accelerometer signal variance to adaptively weight samples for feature training. The results of our five-fold cross-validation experimental evaluation using the PD-BioStampRC21 dataset show that our method achieved an accuracy of 86.67% and an AUC score of 95.56%, representing a significant improvement over the baseline model's accuracy of 72.67% and surpassing other existing methods. Our method enables more accurate detection of subtle tremor patterns in real-world conditions compared to conventional methods. The proposed method has shown potential in the integration of wearable sensors and ensemble learning techniques for the early and accessible diagnosis of Parkinsonian hand tremor, ultimately contributing to improved patient care and treatment.

INDEX TERMS Parkinson's disease, ensemble learning, machine learning, tremor detection.

I. INTRODUCTION

Parkinson's disease (PD) has become one of the fastest-growing neurological diseases in the world [1]. Early

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detection is a crucial step in slowing its progression [2], [3]. PD symptoms are currently assessed through manual clinical examinations, which require patients to perform a series of standardized tasks to provide symptom scores. With the aging population, the number of PD patients is increasing, which

is placing significant pressure on the diagnostic processes of medical institutions. Therefore, to facilitate the timely discovery of early symptoms of Parkinson's in potential patients, there is an urgent need for a simple and feasible detection method that can be integrated into daily life, as this would contribute to rapid treatment and significant symptom relief.

Tremors are the one of the early typical symptoms of PD. To date, most studies involving on the automatic assessment of PD tremors have focused on using wearable sensors during specific tests such as hand tremor assessment at resting state and finger-to-nose assessment [2], [4]. Patients typically need to perform some specialized tests under the doctor's instructions to provoke tremors [5], [6], [7]. These limitations increase the complexity in daily life detection. Our choice of using a single wearable sensor is driven by both practical and performance considerations. While multi-sensor approaches have shown promise in laboratory settings, they face significant challenges in real-world applications due to setup complexity, user compliance issues, and increased data processing requirements. Through careful feature engineering and advanced machine learning techniques, our single-sensor approach achieves comparable or superior performance to multi-sensor systems while maintaining simplicity and usability. In this paper, we propose a method that uses daily life data in an attempt to detect Parkinson's symptoms from behavioral patterns, it involves the use of a single accelerometer worn on the wrist to capture the typical PD tremors.

Comparing with PD rest tremor detection, detecting tremor during walking can help differentiate between Parkinson's tremor and other tremor disorders, such as essential tremors. Studies have shown that rest tremor alone is not always sufficient for an accurate diagnosis, and examining tremors during movement provides additional diagnostic value [6], [8]. Clinical experiments have found that PD tremor intensity is enhanced during walking compared with resting, while the intensity of essential tremor with resting tremor is markedly attenuated during walking [9], [10]. Additionally, walking is an activity that needs to be performed daily and has periodicity, which makes it easier to recognize compared to other behavioral patterns. Therefore, we choose walking as our study behavioral pattern and suggest that assessing tremor during walking is clinically useful for accurate recognition of parkinsonian tremor.

Feature extraction plays a critical role in sensor-based analysis. Several studies have investigated the extraction and utilization of Parkinson related features derived from sensor data. For example, Hossen et al. [11] used wavelet decomposition to estimate the approximate power spectral density (PSD) of accelerometer and surface electromyography (sEMG) signals. Zhou et al. [12] proposed using a weighted frequency Fourier linear combiner (WFLC) to estimate the frequency and amplitude of tremors. Hathaliya et al. [13] used the CNN-based features for PD tremor detection. However, it is still challenging to capture the unique features

of Parkinson's subtle tremor movements. We noticed that in the early stages of Parkinson's, patients typically exhibit tremors with a dominant frequency of about 4-6Hz compared to that of the normal human movement of 0-3 Hz [14], [15]. Therefore, we propose a new frequency domain feature called APF, which represents the average peak frequency. Furthermore, based on the small amplitude characteristic of Parkinson's hand tremors, we propose using a new feature called APIT in the time domain to represent the time difference between sensor peaks.

While existing methods have shown promise in PD classification, they face three critical limitations for real-world PD detection. First, most existing approaches rely heavily on multiple sensors placed across the body, making them impractical for long-term monitoring and reducing patient compliance in daily life settings. Second, existing methods typically require specific test protocols or controlled environments, such as finger-to-nose tests or resting state assessments, which fail to capture the natural variability of PD symptoms during daily activities. Third, conventional feature extraction methods focus primarily on isolated movements or static positions, potentially missing the subtle but distinctive tremor patterns that manifest during dynamic activities like walking. These limitations significantly restrict the applicability and effectiveness of current methods in real-world scenarios where continuous, unobtrusive monitoring is essential for early detection and tracking of PD progression. Our work directly addresses these limitations through three key innovations: (1) utilizing a single wrist-worn sensor during natural walking, (2) introducing 2 unique dynamic features, APIT and APF, specifically designed to capture PD's characteristic rapid, small-amplitude tremors during walking, and (3) employing an adaptive ensemble learning approach that distinctively utilizes movement variance information to enhance detection sensitivity, whereas traditional methods rely on uniform weighting schemes that may miss subtle tremor patterns.

In this paper, we explore the effectiveness of wearable sensor data in real life environment, specifically from a single accelerometer, for early detection of Parkinson's symptoms. Our contributions are as follows:

- We developed an innovative, non-intrusive method utilizing a single wrist-worn accelerometer to detect Parkinson's disease tremors in the real world. This approach eliminates the need for specialized tests and devices, facilitating easier, more accurate and more practical usage for early signs of PD.
- We designed a dynamic scanning mechanism to filter the valid walking segment from whole walking sequence. Then, according to the walking periodicity, we combined the time and frequency domain features of motion data and proposed two new accelerometer-based metrics: Average Peak Interval Time (APIT) and Average Peak Frequency (APF). These metrics effectively capture the unique characteristics of PD tremors, thus enhancing the accuracy of tremor detection.

- We proposed a variance adaptive ensemble learning model. This model utilized an adaptive sample initial weights and achieved 0.8667 accuracy in experiments on the PD-BioStamp-RC21 dataset, significantly outperforming existing machine learning methods.

II. RELATED WORK

The evolution of wearable sensor technology has led to significant advancements in the detection and monitoring of PD, particularly that focusing on hand tremor detection. Commonly used wearable sensors include accelerometers, gyroscopes, surface electromyography sensors, and magnetometer sensors [16], [17], [18], [19]. One of the early methods proposed by Barrantes et al. [5] employed accelerometers to record tremor signals over 30-second intervals at rest and with the arm extended, which allowed for a basic analysis using power spectrum features.

Machine learning (ML) has proven effective in enhancing PD symptom detection, with ensemble learning emerging as one of the most advanced ML approaches due to its ability to combine multiple models for improved accuracy and robustness. Researchers have begun applying these sophisticated ensemble methods to PD detection, though primarily in the domain of voice analysis [20], [21], [22]. For instance, recent studies have used decision tree ensemble methods like AdaBoost and random forest classifiers to analyze imbalanced speech datasets, with AdaBoost achieving F1-scores of 0.903 when combined with information gain feature selection [23], while another study achieved 81.5% diagnostic accuracy by optimizing feature selection for ensemble classifiers such as AdaBoost and Random Forest [24]. However, despite the proven effectiveness of ensemble learning in PD voice analysis, its application to accelerometer-based PD detection remains unexplored.

Researchers have made significant progress in sensor-based PD detection. Lonini et al. [2] demonstrated that ML algorithms applied to wearable sensor data could effectively detect bradykinesia and tremor. Lin et al. [7] successfully used IMU and sEMG sensors to detect upper limb tremor, finding strong correlations between sensor measurements and clinical tremor scores. Battista and Romaniello [25] developed “PD-Watch” for continuous movement monitoring, showing promising UPDRS score correlations. However, these studies were largely confined to controlled laboratory environments, requiring specialized tests under physician supervision. Our work bridges this gap by combining the power of ensemble learning with accelerometer data analysis, proposing a method that leverages everyday walking patterns captured by a single wrist-worn accelerometer.

Researchers have increasingly focused on accelerometer-based approaches for PD detection due to their portability, accuracy, and widespread availability in consumer electronics [26]. Recent studies have demonstrated both the potential and limitations of these approaches. Wang et al. [27] used accelerometers with time and frequency domain features to classify hand tremors using neural networks,

achieving 84.5% accuracy. However, their approach, like many others, relied on specific laboratory based tests that do not reflect real world conditions. Various approaches to dynamic feature extraction have also been proposed for PD detection. Hossen et al. [11] introduced wavelet based features for tremor analysis during specific motor tasks, while Zhou et al. [12] developed frequency domain features for standardized movement assessments. However, these features were primarily designed for controlled movements rather than natural walking patterns. Hathaliya et al. [13] attempted to capture dynamic characteristics during daily activities using multiple sensors, but their approach required complex sensor fusion and showed reduced accuracy of 74.1% in real world settings.

These accelerometer based studies, while showing promise, face several critical limitations in real world applications. First, they predominantly rely on controlled movements or specific tasks that fail to capture natural daily activities. Second, many employ complex multi sensor setups that reduce patient compliance and increase system complexity. Third, their feature extraction methods are typically optimized for laboratory conditions rather than real world variability. Fourth, they show limited consideration of the temporal dynamics of tremor patterns during continuous movement, particularly in natural walking conditions where tremor characteristics may differ significantly from controlled tests.

Our work addresses these limitations through several key innovations. Our VA AdaBoost algorithm uniquely incorporates signal variance information into the ensemble learning process, making it more sensitive to subtle tremor patterns that may be overlooked by conventional uniform weighting schemes. Our APIT and APF features are specifically designed to capture tremor characteristics during walking, addressing limitations of traditional feature sets that often focus on isolated movements or static positions. Furthermore, our approach achieves superior performance with 86.67% accuracy using only a single sensor, making it more practical for real world applications while maintaining high detection accuracy.

In this paper, we focus specifically on the detection of PD tremors during walking using a single wrist-worn accelerometer, gathered through real-world experiments, using the PD-BioStampRC21 dataset collected by Adams et al. [28]. This PD real-world dataset has been used in recent papers [13], [29], [30]. However, these prior studies did not differentiate between various test types and scenarios. To our knowledge, little previous work has focused specifically on developing machine learning models based on motion data collected from real-world environments using accelerometer data, and this paper is the first to focus on the identification of Parkinson's tremors during walking by utilizing wrist accelerometer data.

In our study, we proposed two new accelerometer-based metrics, APIT and APF, for hand tremor intensity and frequency during walking, respectively. These features

TABLE 1. Comparison with related works.

Reference	Sensors Used	Data Collection	Tremor Detection During Walking	Environment Setting	Time Features	Frequency Features	Dynamic Features	Method
Barrantes et al. (2017) [5]	Accelerometer	Rest and arm-extended positions test	X	Laboratory	X	1	X	Statistical analysis
Lonini et al. (2018) [2]	Accelerometer, Gyroscope	UPDRS-Part3-based tasks	O	Representative of common daily tasks	5	5	X	Random Forest, CNN
Lin et al. (2023) [7]	IMU, sEMG	6 clinical action tests	X	Laboratory	11	X	X	FCN+LSTM
Wang et al. (2023) [27]	Accelerometer	Several function experiments	X	Laboratory	7	2	X	BP neural network
Igene et al. (2023) [30]	Accelerometer	Collected in sitting, standing, and walking	Not Specified	Real world	8	8	X	SVM
Hathaliya et al. (2022) [13]	Accelerometer	Not specified	Not Specified	Real world	Not Specified	NotSpecified	X	CNN based on ML-features
Our method	Accelerometer	Collected in Walking	O	Real world	6	5	2 (APIT, APF)	VA-AdaBoost

are distinct from traditional time and frequency-domain features, allowing for more precise tremor characterization during walking, an activity that has been understudied in prior work. Additionally, we utilize an advanced ensemble learning model, VA-AdaBoost, to enhance tremor detection performance by mitigating common ML issues such as overfitting and insufficient complexity capture.

Unlike previous studies, which often focus on controlled laboratory environments, our method leverages real-world activity data to detect PD tremors in natural settings. This makes it more applicable for continuous monitoring of PD symptoms during daily life, offering significant improvements in both detection accuracy and practicality. Table 1 compares our approach with prior studies about PD tremor detection, highlighting key distinctions in terms of sensor type, data collection methods, and environment settings. It also highlights the features extracted from the data and the methods applied.

III. MATERIALS AND METHODS

In this study, we developed a novel, simplified automatic method that uses a wrist accelerometer to assess the presence of PD tremors. In the future, it is expected that this method can be integrated into daily life, with potential patients simply needing to wear a wristband to allow for PD tremor detection while walking without causing any disturbance to their daily routine. Fig. 1 shows the pipeline of the proposed method. First, raw accelerometer data is collected from a single wrist-worn sensor during natural walking activities. Next, the signal undergoes preprocessing with a low-pass filter and a dynamic scanning mechanism to isolate valid walking segments. The third stage involves comprehensive feature extraction, where both traditional features (time and frequency domain) and our novel features (APIT and APF) are computed from the processed data. Finally, these features are fed into our VA-AdaBoost classification model, which employs variance-adaptive sample weighting to distinguish between PD and healthy subjects.

A. GAIT ACCELERATION DATASET

The dataset we utilized, PD-BioStampRC21, was collected by Adams et al. [28], and has been made publicly available on

the IEEE DataPort [31]. Since our research utilized a publicly available dataset, the requirement for informed consent was waived. This study was reviewed and approved by the Institutional Review Board of Sungkyunkwan University (SKKU 2024-03-048). It comprises data from 17 participants with Parkinson's Disease (PD) and 17 Healthy Controls (HC) that was collected using wearable sensors, specifically the lightweight MC 10 BioStamp RC sensors, attached to the trunk, left and right upper anterior thighs, and left and right anterior forearms. Our study exclusively used accelerometer readings taken from the right forearm of participants during walking. This choice was based on both clinical evidence and practical considerations, as the forearm provides reliable tremor signal capture during natural walking movements. The single-sensor approach not only simplifies data collection and processing but also enhances user comfort and compliance, which are critical factors for real-world applications. This aligns with our goal of developing a practical, user-friendly solution for continuous PD monitoring in daily life. The accelerometer records acceleration values in the X, Y, and Z axes at a sampling rate of 31.25 Hz, and it stores these measurements in units of 'g'. Table 2 lists the details of the dataset. Fig. 2 shows the signals of one representative HC sample data and one representative PD sample data, respectively.

TABLE 2. Dataset details.

Characteristic	PD	HC
Number	17	17
Sex	10/7	4/13
Age	66.4[11.3]	64[9.9]
Years since diagnosis	4.8[4.0]	N/A
MDS-UPDRS total motor score	20.9[7.9]	2.2[2.1]

Note: Sex, men/women; a[b], mean[standard deviation]; MDS-UPDRS, Movement Disorder Society-Unified Parkinson's Disease Rating Scale; N/A, Not Applicable.

B. DATA PREPROCESSING

We collected data from right forearm triaxial acceleration measurements in PD and HC participants. First, we checked the data for missing values, after which we excluded two

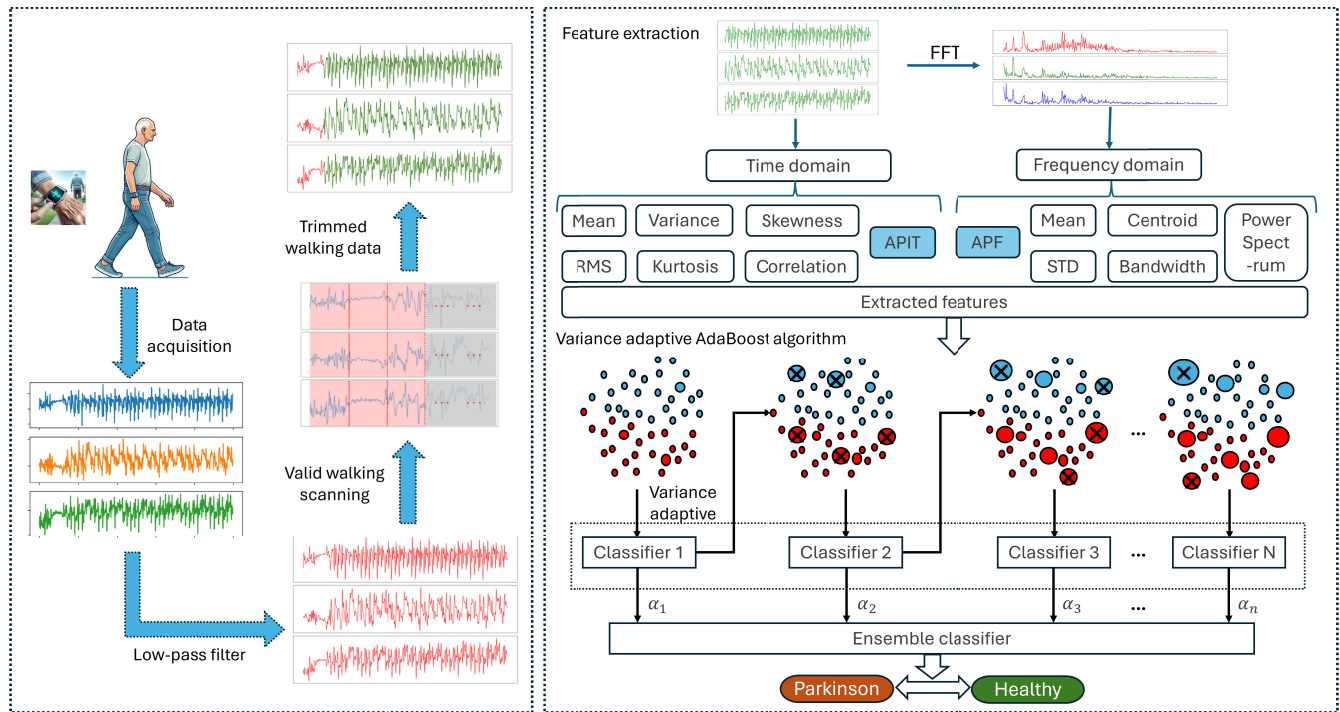


FIGURE 1. Pipeline of the proposed method. APIT, Average Peak Interval Time; APF, Average Peak Frequency. The details will be explained in Section III-D.

healthy subjects and three Parkinson's disease subjects. Next, we applied a Butterworth low-pass filter to the accelerometer data. Our configuration involved a 3rd-order setting, a sampling rate of 31.25 Hz, and a cutoff frequency of 6.5 Hz. Our rationale behind using this filter was to attenuate high-frequency noise while preserving the essential signal characteristics related to human gait patterns. The filtered data from each of the accelerometer's axes (X, Y, Z) was then derived, ultimately providing a cleaner dataset for further analysis.

C. IMPLEMENTATION DETAILS

The proposed model was implemented using the PyTorch framework, with all experiments conducted on an NVIDIA GeForce RTX 2080Ti GPU to ensure computational efficiency. The classification process begins with label encoding, where samples from the PD group are assigned a binary label of 1, and those from the HC group are assigned a label of 0. To maintain class balance and improve the robustness of performance evaluation, a Stratified 5-Fold Cross-Validation strategy is adopted. This method partitions the dataset into five subsets while preserving the proportion of PD and HC samples in each fold, thereby enabling an unbiased assessment of the model's classification performance.

Through extensive experimentation and hyperparameter optimization, we determined the following configurations yielded the best classification results: Support Vector Machine (SVM) achieved optimal performance using a

Radial Basis Function (RBF) kernel with regularization parameter (C) of 1 and gamma value of 0.1; K-Nearest Neighbors (KNN) performed best with $k = 7$; Random Forest (RF) showed peak accuracy using 100 trees and maximum depth of 5; and Logistic Regression (LR) achieved optimal results with C value of 1. For the best-performing Multilayer Perceptron (MLP) architecture consisted of one hidden layer containing 100 neurons, with ReLU activation for the hidden layer and sigmoid for the output layer. The optimal Convolutional Neural Network (CNN) configuration utilized one convolutional layer with 128 filters. For the boosting models, a decision tree with a maximum depth of 1 is utilized as the base learner, with hyperparameters set to a learning rate of 0.1 and 200 estimators.

D. MOVING VARIANCE-ENHANCED SCANNING MECHANISM

There are some inevitable delays when the sensor starts and stops recording, and the sensor can also fall off, alongside other possible issues. Therefore, to preserve only the effective walking part and extract meaningful features in the next step, we proposed a mechanism that could be used to filter out the real walking fragments. This technique involved calculating the moving variance with overlap for each axis, thus capturing local variability in the data more effectively. Then, to focus the analysis on the most relevant fragments of the data, we established a variance threshold, which was set at 10% of the maximum mean-variance observed after the 200th data point. We computed the moving variance over a window of

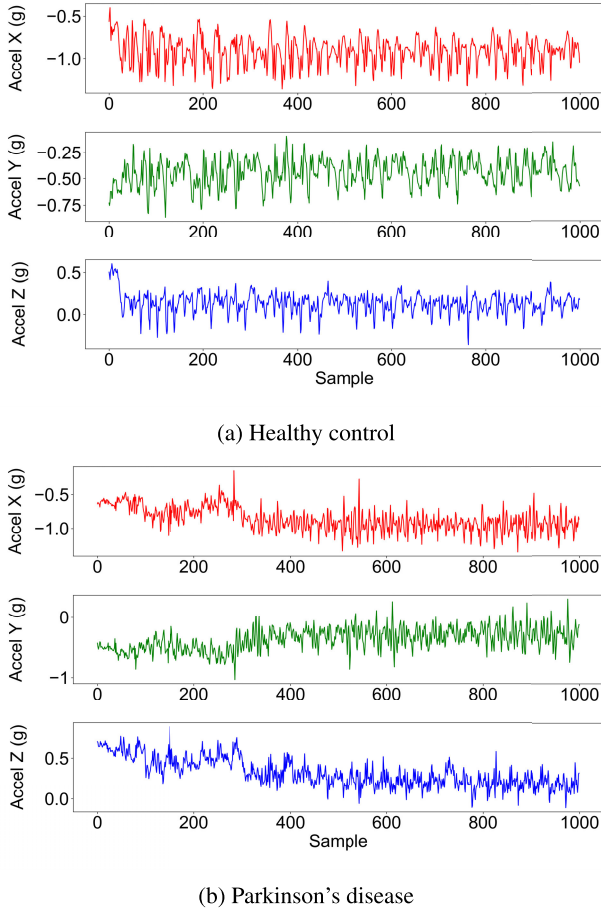


FIGURE 2. The representative participants' raw accelerometer signal. The first 1000 data points are selected for visualization.

64 data points, with an overlap of 10%. Next, we identified the start and end indices where the variance in all three axes simultaneously exceeded this threshold. Fig. 3 shows the moving variance-enhanced scanning mechanism. This approach allowed us to trim the data to a segment comprising significant motion, likely representing the actual gait test period. The data trimmed in this way was now focused on the key segment and therefore ready for the subsequent analyses.

E. FEATURE SELECTION AND EXTRACTION

After obtaining the valid walking segments, to effectively classify individuals, we carefully selected a set of relevant features from the accelerometer data. Each feature was chosen based on its potential ability to capture essential characteristics related to hand tremors and movement during walking. We ultimately chose six-time domain features, five frequency domain features, and two derived features for each of the three axes. Therefore, we obtained 39 features in total, and we used Z-normalization to transform the features into a distribution with a mean of 0 and a standard deviation of 1.

1) TIME DOMAIN FEATURES

The basic features in the time domain are mean, variance, skewness, and kurtosis. Mean represents the average value of the data points, thus providing a baseline for hand movement intensity. Variance captures the variability of movement, which is expected to be higher in Parkinson's disease due to associated tremors. Skewness indicates the asymmetry of the accelerometer signal distribution, potentially highlighting irregular tremor patterns. Moreover, high kurtosis values may signify the presence of pronounced spikes in movement, which are characteristic of tremors. We use $x_{axis,i}$ to represent an accelerometer reading along a particular axis, and N is the number of samples. The mean and variance are expressed as shown in Eq.(1) and Eq.(2), respectively:

$$\mu_{axis} = \frac{1}{N} \sum_{i=1}^N x_{axis,i}. \quad (1)$$

$$\sigma_{axis}^2 = \frac{1}{N} \sum_{i=1}^N (x_{axis,i} - \mu_{axis})^2. \quad (2)$$

a: ROOT MEAN SQUARE (RMS)

RMS provides a measure of the signal's magnitude, which is useful for identifying tremor intensity. Eq.(3) shows the RMS calculation:

$$RMS_{axis} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_{axis,i}^2}. \quad (3)$$

b: CORRELATION COEFFICIENT

The correlation between different axes of the accelerometer can reveal the presence of coordinated or uncoordinated movements. We choose the Pearson correlation coefficient (PCC), which is a correlation coefficient that measures the linear correlation between two sets of data. It can range from -1 to $+1$, with values closer to -1 indicating a strong negative correlation, values closer to $+1$ indicating a strong positive correlation, and values closer to 0 indicating no correlation. Eq.(4) shows the PCC calculation:

$$\rho_{axis1,axis2} = \frac{\sum (x_{axis1} - \mu_{axis1})(x_{axis2} - \mu_{axis2})}{\sqrt{\sum (x_{axis1} - \mu_{axis1})^2 \sum (x_{axis2} - \mu_{axis2})^2}}. \quad (4)$$

2) FREQUENCY DOMAIN FEATURES

We apply the Fast Fourier Transform (FFT) to the acceleration data of each axis to convert the signal from the time domain to the frequency domain. For each $axis \in \{x, y, z\}$, there is a signal $x_{axis}(n)$, where n is an integer that represents the sample index, and the discrete Fast Fourier Transform (DFT_{axis}) is defined by Eq.(5):

$$X_{axis}(k) = \sum_{n=0}^{N-1} x_{axis,n} e^{-i \frac{2\pi}{N} nk} \quad k = 0, 1, \dots, N-1 \quad (5)$$

where N is the total number of samples and k is the frequency index. For a 3-axis accelerometer, we perform the FFT for each axis independently. We assume that the signals for each

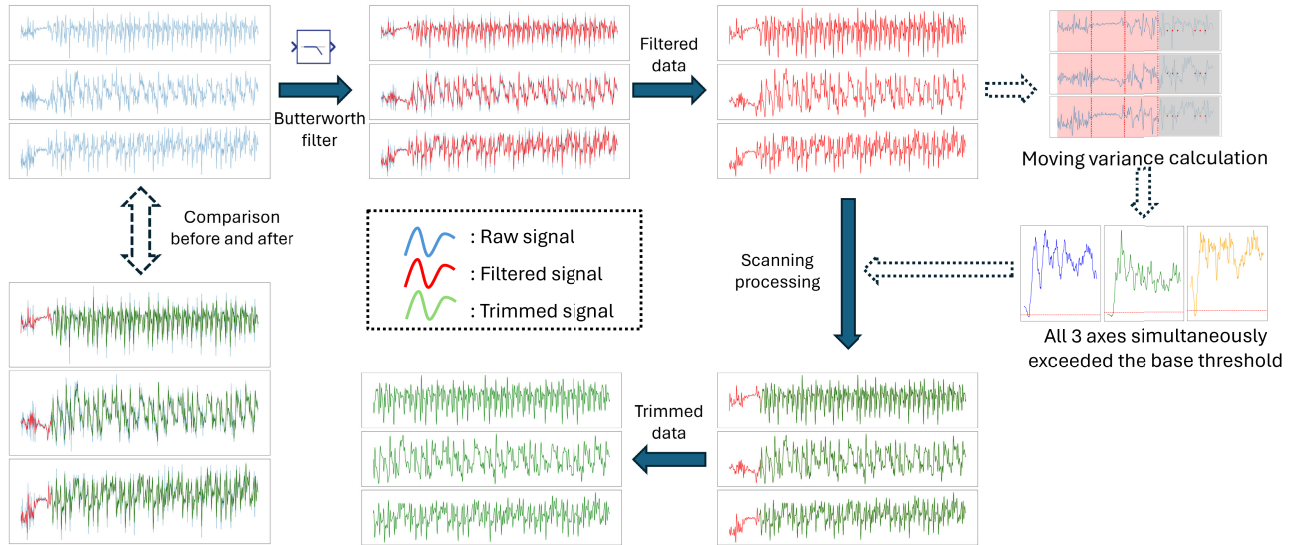


FIGURE 3. Moving variance-enhanced scanning mechanism.

axis are $X_x(n)$, $X_y(n)$, $X_z(n)$. Since the FFT is symmetric around the Nyquist frequency, we only consider the first half $k = 0$ to $N/2$ of the FFT output. In the equation below, we will use N to represent $N/2$.

For basic frequency domain features, we choose mean and standard deviation. The mean value of the FFT helps elucidate the dominant frequencies in hand movement, which are likely to be different in Parkinson's patients. The standard deviation of the FFT is used to capture the variability in the frequency domain, which might be indicative of tremor inconsistency.

a: POWER SPECTRUM

This represents the distribution of power into frequency components, which is crucial for understanding tremor dynamics. After computing the FFT, we typically find the magnitude spectrum using Eq.(6):

$$|X_{axis}(k)| = \sqrt{\text{Re}(X_{axis}(k))^2 + \text{Im}(X_{axis}(k))^2}, \quad (6)$$

where $\text{Re}(X_{axis}(k))$ and $\text{Im}(X_{axis}(k))$ are the real and imaginary parts of the FFT output, respectively. We use Eq.(7) to obtain the power spectrum P_{axis} :

$$P_{axis}(k) = |X_{axis}(k)|^2. \quad (7)$$

b: SPECTRAL CENTROID

The spectral centroid provides a measure of the “center of mass” of the spectrum, which can be interpreted as a kind of average of the frequencies present in the signal, which is weighted by their amplitudes. These values indicate the dominant frequency characteristics of the hand movement that occurs during walking for each axis. Eq.(8) shows the calculation:

$$C_{axis} = \frac{\sum_{k=0}^N k P_{axis}(k)}{\sum_{k=0}^N P_{axis}(k)}. \quad (8)$$

c: SPECTRAL BANDWIDTH

This is a measure that indicates the width of the band of frequencies that contain most of the energy of the signal. It can be used to understand the range of frequencies over which hand movements occur. A larger bandwidth suggests that there is a wider range of frequencies that contribute significantly to the signal's behavior. The calculation of BW_{axis} is as shown in Eq.(9):

$$BW_{axis} = \sqrt{\frac{\sum_{k=0}^N (k - C_{axis})^2 P_{axis}(k)}{\sum_{k=0}^N P_{axis}(k)}}. \quad (9)$$

We visualize the spectral centroid and spectral bandwidth using the PD and HC sample data, as shown in Fig. 4 and Fig. 5. For the spectral centroid, we notice that the PD sample generally has higher values, particularly on the Y-axis. A higher centroid value suggests that the motion in that axis has a higher average frequency component, potentially indicating the presence of faster or more dynamic movements. As can be seen in Fig. 5, the Y-axis in particular shows a much larger bandwidth compared to the X and Z axes, thus indicating a greater diversity of frequency components in the vertical movement captured in the gait data.

3) PEAK DYNAMIC FEATURES

We proposed two derived peak dynamic features for each of the three axes, i.e., APIT and APF.

a: AVERAGE PEAK INTERVAL TIME (APIT)

Since Parkinson's patients are likely to have a small magnitude of hand movement, the interval between peaks can help differentiate rhythmic tremor patterns from normal hand motions. To describe this, we propose a new feature called Average Peak Interval Time (APIT), as shown in

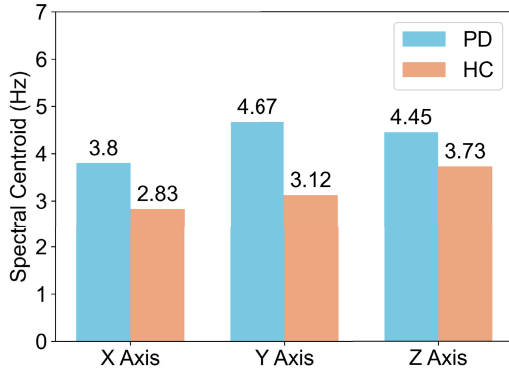


FIGURE 4. Comparison of spectral centroid between Parkinson's Disease (PD) and Healthy Control (HC) subjects. PD shows consistently higher centroid values across all axes, with the most pronounced difference in the Y-axis (4.67 Hz vs 3.12 Hz), indicating higher frequency components in PD movements.

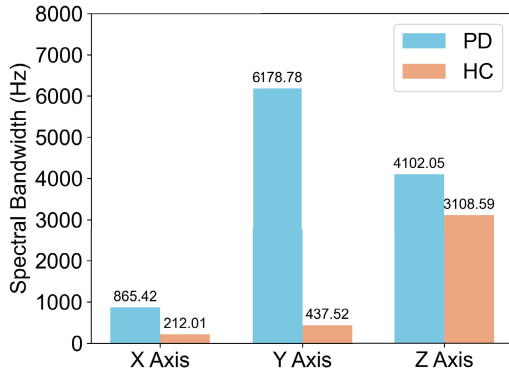


FIGURE 5. Comparison of spectral bandwidth between Parkinson's Disease (PD) and Healthy Control (HC) subjects. PD exhibits notably larger bandwidth values, particularly in the Y-axis (6178.78 Hz vs 437.52 Hz), suggesting a wider range of frequency components in tremor movements.

Eq.(10) and Eq.(11). APIT measures the typical time between successive peaks in hand movement, essentially capturing how frequently someone's hand shows repeated movements. Shorter intervals often indicate the rapid, rhythmic tremors characteristic of Parkinson's disease. First, the time intervals between consecutive peaks are calculated as:

$$\Delta t_{peaks_{axis,i}} = t_{peaks_{axis,i}} - t_{peaks_{axis,i-1}}, \quad (10)$$

where $t_{peaks_{axis,i}}$ and $t_{peaks_{axis,i-1}}$ are the timestamps of the i -th peak and the previous peak, respectively.

Next, the APIT is defined as:

$$APIT_{axis} = \frac{\sum_{i=2}^{N_{peaks_{axis}}} \Delta t_{peaks_{axis,i}}}{N_{peaks_{axis}} - 1}, \quad (11)$$

where $N_{peaks_{axis}}$ is the total number of peaks identified in the signal.

In our analysis, for peak detection within the signal data, we identify local maxima in the signal by comparing each data point with its immediate neighbors, which effectively

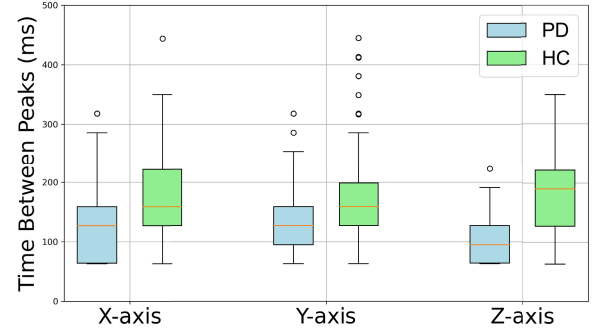


FIGURE 6. Comparison of APIT between Parkinson's disease (PD) and healthy control (HC) subjects. The boxplots show lower APIT values across all axes for PD patients (blue) compared to healthy controls (green), indicating shorter time intervals between peaks in the acceleration signal. The consistently shorter intervals in PD patients reflect the characteristic rapid, rhythmic tremors of Parkinson's disease compared to the more varied and slower movement patterns in healthy individuals.

captures any point that is higher than the points directly before and after it. We use the boxplot to visualize the distribution of time intervals between successive peaks for each axis, as shown in Fig. 6. From the results, we note that PD patients generally have lower values than the control group, indicating that they have lower APIT and shorter time intervals between peaks. This visualization can help elucidate the characteristics of the movement captured by the accelerometer in terms of its regularity and variability across the three axes.

b: AVERAGE PEAK FREQUENCY (APF)

We have observed that patients with Parkinson's disease exhibit high peak frequency values [14], [15]. The peaks in the frequency spectrum represent the frequencies with the highest amplitudes, which indicate the frequencies that are most dominant in the motion recorded. In our proposed method, we identified the average frequency at which peaks occur, which could be indicative of tremor frequency. When finding the peaks in the spectrum, we could set a threshold to find the peaks.

Fig. 7 and Fig. 8 respectively show the FFT peaks for the control sample and the PD sample when we set the $threshold = 20$. We can notice that Parkinson's patients have a higher frequency, so we propose a new feature called average peak frequency APF_{axis} to represent this feature, as shown in Eq.(12).

$$APF_{axis} = \frac{\sum_{i \in peaks} F_{axis,i}}{N_{peaks_{axis}}}, \quad (12)$$

where $F_{axis,i}$ is the frequencies at the peaks of the magnitude spectrum and $N_{peaks_{axis}}$ is the number of peaks identified. The peaks in the magnitude spectrum can be identified using a specified threshold.

APF represents the mean frequency of dominant oscillatory components in the movement signal after Fast Fourier

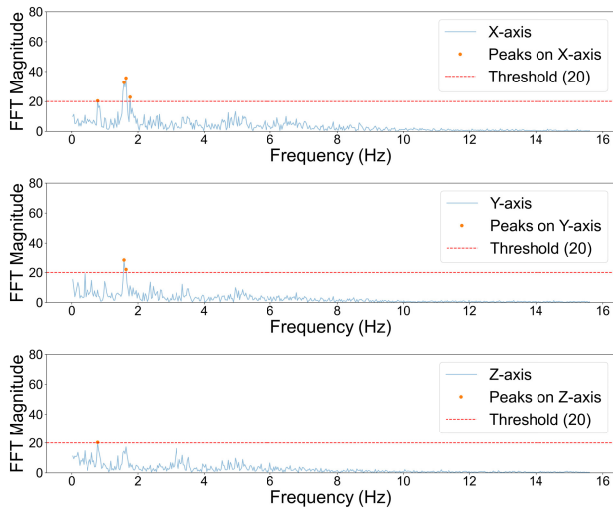


FIGURE 7. A representative HC participant's FFT peaks. The frequency spectrum shows dominant peaks primarily in the lower frequency range (0-3 Hz), consistent with normal physiological movements during walking. A threshold of 20 (red dashed line) is used for peak detection.

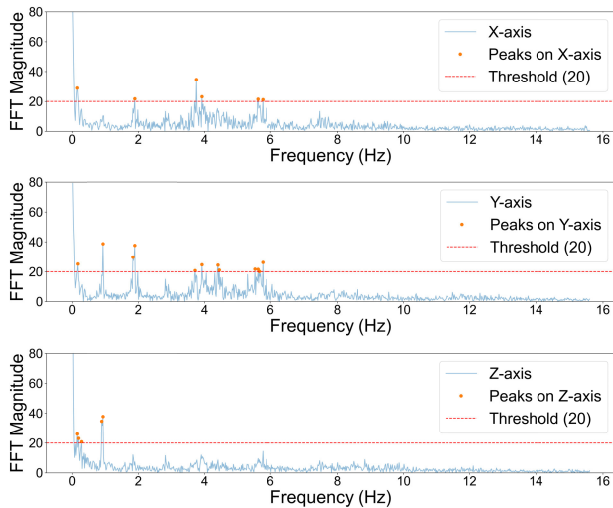


FIGURE 8. A representative PD participant's FFT peaks. Multiple peaks are observed in higher frequency ranges (4-6 Hz), particularly visible in the Y-axis, demonstrating the characteristic higher-frequency components of PD tremors. A threshold of 20 (red dashed line) is used for peak detection.

Transform (FFT). Unlike a simple count of hand movements per second, APF specifically captures the frequency characteristics of movement by identifying peaks in the frequency spectrum above a certain threshold. In our study, PD patients typically showed higher APF values (concentrated in the 4-6 Hz range) compared to healthy controls, consistent with clinical observations that Parkinsonian tremors manifest at higher frequencies than normal physiological movements (which typically occur at 0-3 Hz). This frequency-domain feature effectively distinguishes the characteristic higher-frequency components of PD tremors from normal movement patterns during walking.

For a more intuitive understanding, if we were to look at the frequency spectrum of hand movements during walking, healthy individuals show dominant movement patterns at lower frequencies (like the natural arm swing during walking), while PD patients show additional significant movement patterns at higher frequencies (representing tremors superimposed on the walking movement). APF quantifies this by averaging the frequencies where we see peaks in the spectrum, effectively capturing the higher-frequency tremor components characteristic of PD.

F. ENSEMBLE LEARNING METHOD

Accelerometers are used to capture three-dimensional motion data $\mathbf{A} = \{a_x, a_y, a_z\}$, where a_x , a_y , and a_z are the accelerations along the x, y, and z axes, respectively. We then employ the feature extraction function, denoted as $f(\cdot)$, to the time-series accelerometer data to derive time domain features, frequency domain features, and peak dynamic features, as described in Section III-D. This process transforms the raw data into a feature vector $\mathbf{x} = f(\mathbf{A})$. For training our model, we aggregate individual feature vectors into a feature matrix. Thus, $\mathbf{X}$ represents our training set, comprising N examples: $\mathbf{X} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\}$, where each $\mathbf{x}_i$ is an individual feature vector derived from a sample and y_i its corresponding class label.

To address the challenge of accurately classifying gait patterns from accelerometer data, where subtle nuances are crucial for distinguishing healthy individuals from those with PD, we propose the Variance Adaptive AdaBoost (VA-AdaBoost) algorithm. Our approach automatically adjusts its learning process based on the degree of movement variation detected in the sensor data. By assigning greater weight to movement patterns that exhibit significant variation, which often signals tremor activity, our algorithm more effectively distinguishes normal gait fluctuations from tremor-specific patterns. Traditional AdaBoost may overlook these subtle variations because it initializes sample weights uniformly, potentially missing highly informative samples that display distinctive variance. By adapting the initial weighting scheme to reflect the observed accelerometer variance, VA-AdaBoost enhances sensitivity to the critical gait characteristics intrinsic to PD.

Specifically, for each training sample (x_i, y_i) , where x_i is the feature vector and y_i the corresponding label, we compute the variance V_{A_i} from the raw accelerometer data $\mathbf{A}_i = \{a_{x,i}, a_{y,i}, a_{z,i}\}$:

$$V_{A_i} = \frac{1}{3} \sum_{\text{axis} \in \{x, y, z\}} \left[\frac{1}{T} \sum_{t=1}^T (a_{\text{axis},i}(t) - \bar{a}_{\text{axis},i})^2 \right], \quad (13)$$

where T is the number of time points, $a_{\text{axis},i}(t)$ is the raw acceleration along each axis at time t , and $\bar{a}_{\text{axis},i}$ is the mean acceleration for the axis over the time period T . The initial weight of each sample is then assigned proportionally to its

variance:

$$w_{i,1} = \frac{V_{A_i}}{\sum_{j=1}^N V_{A_j}}. \quad (14)$$

This adaptive initialization enhances the focus on more informative samples from the initial iteration. The algorithm then proceeds through T rounds of training. In each iteration t , a weak learner h_t is trained using the current sample weights $w_{i,t}$. The error of the weak learner h_t is calculated as the weighted sum of misclassified samples:

$$\epsilon_t = \sum_{i=1}^N w_{i,t} (y_i \neq h_t(x_i)). \quad (15)$$

Based on this error, the weight of the weak learner is computed as Eq. (16), which reflects its contribution to the final model.

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right). \quad (16)$$

The sample weights are then updated to emphasize misclassified instances, allowing subsequent learners to focus on these difficult cases as Eq. (17). To maintain a proper distribution, the sample weights are then normalized using Eq. (18)

$$w_{i,t+1} = w_{i,t} \cdot \exp(-\alpha_t y_i h_t(x_i)). \quad (17)$$

$$w_{i,t+1} = \frac{w_{i,t+1}}{\sum_{j=1}^N w_{j,t+1}}. \quad (18)$$

After T iterations, the weak learners are combined into a final model, where the output is a weighted majority vote:

$$H(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right). \quad (19)$$

By integrating the variance of accelerometer readings into the weight initialization and focusing on misclassified instances, the VA-AdaBoost algorithm not only iteratively updates the weights and combines weak learners but also prioritizes the most informative samples from the outset.

Stacking is also a popular ensemble learning approach that differs from boosting, and it employs a triad of machine learning models as base learners. Each base learner $h_i(\cdot)$ is trained on the feature set $\mathbf{X}$ to generate predictions p_i for the target variable Y : $p_i = h_i(\mathbf{X})$, $i \in \{1, 2, \dots, n\}$, where $h_i(\cdot)$ represent base learners. Then, the predictions p_i from the aforementioned base learners are aggregated by a meta-learner through an ensemble method. This ensemble employs soft voting, which considers the predicted probabilities from the base learners to compute a weighted average prediction. The weights are determined based on the performance of each model during the validation phase. The ensemble learning model can be expressed by Eq.(20).

$$P_{\text{ensemble}}(Y) = \frac{\sum_{i=1}^3 w_i \cdot P(Y|h_i(\mathbf{X}))}{\sum_{i=1}^3 w_i}, \quad (20)$$

where w_i is the weight for the i -th model's prediction, which is derived from its cross-validated performance, and $P(Y|h_i(\mathbf{X}))$ is the predicted probability of the target variable Y given the feature set $\mathbf{X}$ and the i -th model.

The final prediction $\hat{Y}$ is then obtained by selecting the class with the highest ensemble probability: $\hat{Y} = \arg \max P_{\text{ensemble}}(Y = c)$, $c \in \{1, \dots, K\}$, where K is the number of classes.

G. EVALUATION METRICS

We choose accuracy, precision, sensitivity, F1 score, Area Under the Curve (AUC) and Matthews Correlation Coefficient (MCC) to evaluate the performance of our method. Accuracy reflects the overall effectiveness of the model, which is calculated as the ratio of correctly identified observations (both Parkinson's patients and healthy individuals) to the total number of observations as $\text{Accuracy} = (TP + TN)/(TP + TN + FP + FN)$. Precision measures the model's ability to correctly identify Parkinson's patients among all cases labeled as such, thus highlighting the reliability of positive predictions as $\text{Precision} = (TP)/(TP + FP)$. Sensitivity, or recall, is a particularly critical consideration in a medical context. It quantifies the model's capacity to correctly identify Parkinson's patients from all actual cases, thus ensuring minimal missed diagnoses as $\text{Sensitivity} = (TP)/(TP + FN)$. The F1 score is the harmonic mean of precision and sensitivity, and it serves as a single score that balances both false positives (precision) and false negatives (sensitivity). It is calculated using the following equation:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}. \quad (21)$$

The AUC value quantifies the overall ability of the model to discriminate between participants with and without Parkinson's disease. An AUC of 1 indicates perfect discrimination, while an AUC of 0.5 suggests no discriminative power, i.e., an outcome equivalent to random guessing. AUC is particularly important in medical applications because it provides an aggregate measure of performance across all possible classification thresholds. This characteristic is beneficial in medical diagnostics, where the threshold for classifying a patient as having Parkinson's might vary based on clinical considerations.

dataset. The MCC is calculated using the following Equation: Moreover, given the dataset's limitations, we selected the MCC as a balanced metric. It is a robust statistical rate that yields a high score only if the prediction performs well across all four categories of the confusion matrix (true positives, false negatives, true negatives, and false positives). This performance must be proportional to both the size of the positive and negative elements in the dataset. The MCC is calculated using the following equation:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \quad (22)$$

TABLE 3. Comparison with other machine learning methods. The best results are highlighted in bold.

Model	Accuracy	Precision	Sensitivity	F1 score	AUC	MCC
SVM	0.5867 ± 0.07	0.3333 ± 0.42	0.2000 ± 0.27	0.2333 ± 0.29	0.3333 ± 0.26	0.1561 ± 0.19
KNN	0.6267 ± 0.16	0.5000 ± 0.45	0.3667 ± 0.31	0.4076 ± 0.34	0.6944 ± 0.15	0.2639 ± 0.32
RF	0.7600 ± 0.17	0.6833 ± 0.37	0.6333 ± 0.37	0.6381 ± 0.34	0.8222 ± 0.13	0.5306 ± 0.34
LR	0.6933 ± 0.12	0.7833 ± 0.19	0.5667 ± 0.25	0.6181 ± 0.16	0.6111 ± 0.22	0.4200 ± 0.24
MLP	0.6267 ± 0.11	0.7200 ± 0.23	0.5667 ± 0.25	0.5776 ± 0.12	0.6111 ± 0.23	0.3014 ± 0.25
CNN	0.7267 ± 0.07	0.7200 ± 0.14	0.8000 ± 0.16	0.7367 ± 0.06	0.6222 ± 0.15	0.4975 ± 0.16
MTSCM	0.7267 ± 0.22	0.7867 ± 0.27	0.7000 ± 0.27	0.7100 ± 0.22	0.7278 ± 0.22	0.4867 ± 0.45
AdaBoost	0.8333 ± 0.15	0.8700 ± 0.17	0.8667 ± 0.27	0.8214 ± 0.19	0.8111 ± 0.25	0.7203 ± 0.24
Proposed method	0.8667 ± 0.12	0.9000 ± 0.12	0.8667 ± 0.27	0.8429 ± 0.18	0.9556 ± 0.09	0.7723 ± 0.20

MCC ranges from -1 to $+1$, where $+1$ indicates perfect prediction, 0 equals random chance, and -1 signifies complete disagreement between prediction and observation.

IV. RESULTS AND DISCUSSION

In this study, we evaluated the performance of our proposed ensemble learning model using a 5-fold cross-validation approach. This method ensures a robust evaluation by dividing the dataset into five subsets, where each subset is used once as a test set while the remaining subsets form the training set. This process mitigates the risk of overfitting and provides a more generalized performance metric.

A. PERFORMANCE OF THE ENSEMBLE LEARNING MODEL

To illustrate the effectiveness of our model, we implemented and compared its performance with those of four typical machine learning methods, including Support Vector Machine (SVM) [32], K-Nearest Neighbors (KNN) [33], Random Forest (RF) [34], Logistic Regression (LR) [35], and three deep learning methods including Multilayer Perceptron (MLP) [36], Convolutional Neural Network (CNN) [37] and Multivariable Time-Series Classification Model (MTSCM) [7]. MTSCM employs fully convolutional networks and a long short-term memory network to recognize tremor actions. The comparative results are presented in Table 3.

From the results presented in Table 3, it is evident that our proposed method outperforms the other models in all evaluated metrics. Notably, it achieves significant improvements, with an accuracy of 0.8667 ± 0.12 , a precision of 0.9 ± 0.12 , and a sensitivity of 0.8667 ± 0.27 , comparing to the other models. This indicates that our model has a superior ability of our model to correctly classify instances and is effective in identifying positive cases.

The F1 score and MCC of our model also stand out, with respective values of 0.8429 ± 0.18 and 0.7723 ± 0.20 . These metrics, which consider both precision and sensitivity, further emphasize the balanced performance of our model in terms of both false positives and false negatives. Moreover, the AUC for our method is 0.9556 ± 0.09 , which is significantly higher than that of the baseline models. This particularly high AUC value demonstrates the model's excellent capability in distinguishing between the classes.

We further evaluated the efficacy of our proposed ensemble learning method against other well-established ensemble techniques, including various forms of stacking and boosting methods, which are known for their robust performance in complex machine learning tasks.

We conducted experiments with two different stacking configurations [38]. The first configuration, referred to as Stacking-1, combines K-Nearest Neighbors (KNN), Decision Tree (DT), and Logistic Regression (LR) in its ensemble. The second configuration, Stacking-2, employs KNN, Support Vector Machine (SVM), and LR. For boosting methods, we used three variations: XGBoost [39], GradientBoost [40], and AdaBoost [41]. Table 4 presents the results of these experiments.

The results presented in Table 4 indicate that our proposed method, variance adaptive AdaBoost (VA-AdaBoost), outshines other ensemble methods in several key performance metrics. It achieves the highest values with an accuracy of 0.8667 ± 0.12 , a F1 score of 0.8429 ± 0.18 , and a MCC of 0.7723 ± 0.20 . These metrics are critical in evaluating the overall effectiveness and reliability of the model in a real-world scenario. Stacking-1 shows promising results with an accuracy of 0.7933 ± 0.12 and a high precision rate (0.92 ± 0.16). However, it falls short in terms of F1 score and MCC compared to VA-AdaBoost. On the other hand, Stacking-2 demonstrates lower performance across all metrics, indicating that the choice of models in the stacking process is crucial for optimal performance. Among the boosting methods, GradientBoost exhibits strong performance, particularly in terms of accuracy (0.8333 ± 0.15) and F1 score (0.81 ± 0.19), and it is closely followed by AdaBoost and XGBoost. These results underscore the effectiveness of these boosting techniques in ensemble learning but also highlight the superior performance of our VA-AdaBoost approach.

The error values reported in Tables 3 and 4 represent the standard deviations obtained from our five fold cross validation process. In each fold, the dataset was split into different training and testing sets, with each sample appearing in the test set exactly once across all folds. Traditional machine learning methods such as SVM and KNN show relatively large variations in their performance metrics,

TABLE 4. Comparison with other ensemble learning methods. The best results are highlighted in bold.

Model	Accuracy	Precision	Sensitivity	F1 score	AUC	MCC
Stacking-1	0.7933 $\pm$ 0.12	0.9200 $\pm$ 0.16	0.7000 $\pm$ 0.27	0.7433 $\pm$ 0.16	0.8889 $\pm$ 0.14	0.6428 $\pm$ 0.20
Stacking-2	0.6933 $\pm$ 0.16	0.6500 $\pm$ 0.37	0.5000 $\pm$ 0.33	0.5448 $\pm$ 0.31	0.6333 $\pm$ 0.24	0.4053 $\pm$ 0.33
XGBoost	0.8000 $\pm$ 0.12	0.9500 $\pm$ 0.10	0.6667 $\pm$ 0.3	0.7314 $\pm$ 0.20	0.8667 $\pm$ 0.14	0.6617 $\pm$ 0.20
GradientBoost	0.8333 $\pm$ 0.15	0.9200 $\pm$ 0.16	0.8000 $\pm$ 0.27	0.8100 $\pm$ 0.19	0.8778 $\pm$ 0.19	0.7203 $\pm$ 0.24
AdaBoost	0.8333 $\pm$ 0.15	0.8700 $\pm$ 0.17	0.8667 $\pm$ 0.27	0.8214 $\pm$ 0.19	0.8111 $\pm$ 0.25	0.7203 $\pm$ 0.24
VA-AdaBoost	0.8667 $\pm$ 0.12	0.9000 $\pm$ 0.12	0.8667 $\pm$ 0.27	0.8429 $\pm$ 0.18	0.9556 $\pm$ 0.09	0.7723 $\pm$ 0.20

particularly in precision and sensitivity. This suggests these methods are more sensitive to the composition of training and testing sets. Deep learning approaches including CNN and MTSCM demonstrate moderate variations, likely due to their ability to learn more robust features, though they still show notable fluctuations in sensitivity measures. In contrast, ensemble learning methods generally exhibit more stable performance across different folds. Among these, our proposed VA AdaBoost method shows notably smaller variations in both accuracy and AUC scores compared to other ensemble approaches, while maintaining consistent performance in sensitivity and precision. This reduced variability, particularly evident in the AUC standard deviation, suggests that our method achieves not only higher absolute performance but also more reliable and consistent detection capabilities across different patient subsets. These variations across different methods primarily occur due to the inherent variability in tremor manifestations among different patients, where different training test splits can contain varying distributions of symptom presentations and severity levels.

For comparison with the original AdaBoost, our proposed method outperforms the original one, with improvements in accuracy from 0.8333 to 0.8667, precision from 0.87 to 0.9, and AUC from 0.8111 to 0.9556. Additionally, we present the AUC-ROC curve in Fig. 9. Our proposed method shows a superior model performance with a mean AUC of 0.95, indicating that the model has a high true positive rate and maintains a low false positive rate across all thresholds.

While our method achieves strong performance metrics, the implications of misclassification should be considered. Sensitivity is particularly relevant for evaluating misclassification impact because it directly measures the model's ability to avoid false negatives, which are the most concerning type of error in medical diagnosis as they represent missed disease cases. Our sensitivity of 86.67% is the highest among comparable approaches, meaning our method correctly identifies approximately 87 out of 100 PD cases, though this still means about 13 cases might initially go undetected, potentially delaying early intervention opportunities. However, the high AUC score of 0.9556 demonstrates excellent overall discriminative ability between PD and non PD cases, suggesting the model's reliability as a supportive screening tool for clinical diagnosis.

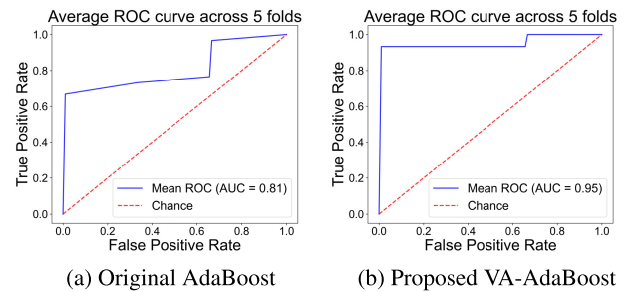


FIGURE 9. ROC comparison between original AdaBoost and our proposed variance adaptive AdaBoost. Our VA-AdaBoost shows superior performance with a mean AUC of 0.95 compared to 0.81 for original AdaBoost, demonstrating better discrimination between PD and healthy subjects across different classification thresholds.

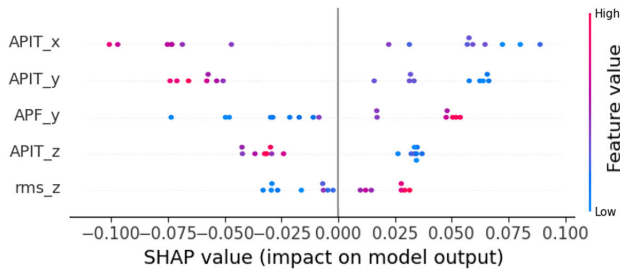
B. EFFECT OF THE PEAK DYNAMIC FEATURES

Our analytical efforts aim to understand the influence of various sensor-derived features on the predictive accuracy of a model tasked with identifying Parkinson's disease patients. We first conducted statistical analyses comparing PD and healthy control groups to validate the discriminative power of our proposed features. Statistical analysis of the $APIT$, APF features, as shown in Table 5, revealed that $APIT_Y$ demonstrated statistically significant differences ($p = 0.0049$) between Parkinson's Disease and healthy control groups, with a negative correlation ($r = -0.5081$). All $APIT$ features showed negative correlations, indicating that PD patients exhibit shorter time intervals between movement peaks, consistent with the characteristic rapid, repetitive nature of parkinsonian tremors. Conversely, APF features displayed positive correlations, reflecting the higher frequency components typical of PD tremors compared to normal physiological movements. $APIT_X$ and $APIT_Z$, along with APF_Y , showed trending significance differences between groups ($0.05 < p < 0.10$).

Additionally, we utilized SHAP (SHapley Additive exPlanations) values, which provide a powerful framework for interpreting machine learning models by quantifying each feature's contribution to the prediction. The SHAP summary plot, as shown in Fig. 10, is confined to the top five features that predominantly impact the model's predictions for Parkinson's disease. The y-axis enumerates the features, while the x-axis measures the SHAP values' magnitude, thus conveying each feature's influence on the prediction.

TABLE 5. Statistical analysis of the APIT and APF.

Feature	Correlation	P-value
$APIT_X$	-0.3447	0.0671
$APIT_Y$	-0.5081	0.0049
$APIT_Z$	-0.3657	0.0511
APF_X	0.0699	0.7186
APF_Y	0.3303	0.0801
APF_Z	0.1873	0.3307

**FIGURE 10.** SHAP visualization. It shows the top five features' impact on PD prediction. Higher APIT values (in red) negatively correlate with PD likelihood (negative SHAP values), while higher APF values positively correlate with PD presence (positive SHAP values), particularly in the Y-axis measurements.

outcome. The color coding—ranging from red to blue—indicates the relative value of the feature, with red signifying higher feature values. Our analysis reveals patterns that strongly align with clinical understanding of PD tremors. There is a direct correlation between higher values of APF features, particularly APF_Y , and the increased likelihood of Parkinson's disease. This finding corresponds with clinical observations that PD tremors typically manifest at higher frequencies during movement [14], [15], particularly during active motions like walking. Conversely, lower values of $APIT$ features, particularly of $APIT_X$ and $APIT_Y$, are indicative of a greater probability of the disease. The negative SHAP values associated with these features reflect their inverse relationship with the prediction of Parkinson's disease, supporting clinical observations that PD patients exhibit more frequent but smaller amplitude movements compared to healthy individuals. This characteristic maps directly to the rapid oscillatory nature of parkinsonian tremors. These machine learning insights not only validate known clinical observations but also suggest that vertical hand movements during walking, captured by our Y-axis measurements, may be particularly diagnostic of PD tremors.

We also evaluated the impact of the two peak dynamic features in the VA-AdaBoost algorithm, with the results depicted in Table 6. The comparison of model performance with and without these features reveals significant improvements across all metrics when the features are included. In particular, accuracy improves from 0.66 to 0.8667, while AUC increases from 0.5556 to 0.9556. Similarly, there are notable enhancements in precision, sensitivity, F1 score, and MCC. These findings demonstrate that APTI and APF two

features can significantly enhance the predictive capabilities of the VA-AdaBoost model.

C. COMPARISON WITH PREVIOUS WORK

Our study introduces a novel ensemble learning approach for detecting Parkinson's disease (PD) hand tremors using data from a single wrist-worn sensor. The proposed methodology represents a significant advancement over existing research by simplifying the detection process, a contrast highlighted in Table 7. In contrast to studies such as Hathaliya et al. [29] and Igene et al. [30], which employ accelerometers in various body locations, our approach only requires a wrist-worn device, thereby enhancing usability and comfort for patients.

Recent studies have attempted various approaches to PD detection using wearable accelerometers, each with distinct limitations. Hathaliya et al. [29] achieved 74.1% accuracy using five accelerometers, but their approach required precise sensor placement and synchronization across multiple body locations, making it impractical for long-term monitoring. Igene et al. [30] improved accuracy to 94.4% using the same sensor configuration, but their method relied on combining data from multiple activities such as sitting, standing, and walking, increasing complexity and processing requirements. While their accuracy was higher, our method achieves comparable diagnostic capability with an AUC score of 0.9556 compared to 0.92, with significantly reduced hardware requirements and implementation complexity.

Our optimized ensemble learning model, incorporating novel accelerometer-based features like Average Peak Interval Time (APIT) and Average Peak Frequency (APF), demonstrates a commendable accuracy of 86.67% along with an outstanding AUC score of 0.9556. This performance is particularly significant when compared to Igene et al. [30]. While their research achieved a marginally higher accuracy rate of 94.4%, our model's performance is distinguished by its superior AUC score of 0.9556 compared to their 0.92. To put this in practical terms, if the model analyzed 100 people's movements, it would correctly classify approximately 96 of them. The AUC score, which is indicative of both sensitivity and specificity, is a critical measure in medical diagnostics. It reflects the model's robustness in accurately distinguishing between PD and non-PD cases, thereby significantly reducing the likelihood of false positives and negatives. This attribute is particularly important in clinical settings where accurate diagnosis is paramount. The higher AUC score of our approach not only highlights its advanced capabilities but also cements its superiority, thus showing that it is a more reliable and effective tool for clinical applications and ongoing patient monitoring in the challenging domain of PD detection.

The limitations of existing approaches become more apparent when examining studies like Lin et al. [42] and Varghese et al. [43], which focused on controlled movement tasks and achieved accuracies of 84.0% and 78.9% respectively. While their use of prescribed movements and multiple sensors might provide cleaner data, such approaches

TABLE 6. Analysis about peak dynamic features.

Model	Accuracy	Precision	Sensitivity	F1 score	AUC	MCC
VA-AdaBoost-w/o	0.6600 ± 0.21	0.7000 ± 0.24	0.5667 ± 0.25	0.6076 ± 0.22	0.5556 ± 0.26	0.3225 ± 0.41
VA-AdaBoost-w	0.8667 ± 0.12	0.9000 ± 0.12	0.8667 ± 0.27	0.8429 ± 0.18	0.9556 ± 0.09	0.7723 ± 0.20

TABLE 7. Comparison with previous works.

	Dataset	Task/Placement	Sensors(# of sensors)	Accuracy	AUC
Hathaliya et al. (2022) [29]	PD-BioStampRC-21	Sitting, Standing, and Walking / Chest, Arm, and Thigh	Accelerometer(5)	0.741	-
Igene et al. (2023) [30]	PD-BioStampRC-21	Sitting, Standing, and Walking / Chest, Arm, and Thigh	Accelerometer(5)	0.944	0.92
Lin et al. (2023) [42]	Collected 164 subjects	TUG test / wrist, Chest, Waist, Thigh, Shank	IMU(10)	0.840	0.91
Varghese et al. (2024) [43]	PADS dataset	14 movement tasks / Wrist	Accelerometer(2)	0.789	-
Our study	PD-BioStampRC-21	Walking / Wrist	Accelerometer(1)	0.867	0.95

fail to capture the natural variability of PD symptoms during daily activities. Lin et al. [42] relied on inertial measurement units in a Timed-Up-and-Go test, while Varghese et al. [43] required participants to perform 14 specific motor assessment tests. In contrast, our method's ability to maintain high accuracy during natural walking represents a significant advancement in practical PD monitoring, enabling continuous assessment without disrupting patients' daily routines.

The superior performance of our single-sensor approach compared to these multi-sensor systems challenges the conventional assumption that more sensors necessarily lead to better detection. Our results suggest that carefully designed features and adaptive learning algorithms can extract more diagnostic value from a single optimally placed sensor than from multiple sensors using conventional analysis methods. This finding has important implications for the future development of wearable PD monitoring systems, suggesting that emphasis should be placed on sophisticated analysis of targeted measurements rather than comprehensive but complex sensor networks. The higher AUC score of our approach not only highlights its advanced capabilities but also demonstrates its superiority as a reliable and effective tool for clinical applications and ongoing patient monitoring in the challenging domain of PD detection.

D. FUTURE WORK

Our study introduces a novel ensemble learning approach for detecting Parkinson's hand tremors using wrist worn sensors, and it shows promising results. The model's accuracy could potentially be enhanced through several approaches. First, expanding the dataset size and diversity would help capture more varied manifestations of PD tremors, improving the model's generalization capabilities. Second, incorporating

temporal dependencies through sequence modeling could better capture the evolution of tremor patterns during extended walking periods. Third, integrating additional contextual information such as walking speed and gait patterns could provide complementary diagnostic signals. Furthermore, implementing an uncertainty quantification mechanism could help identify cases where the model's predictions may be less reliable, prompting more thorough clinical evaluation. Additionally, addressing potential dataset biases is crucial for future development. Our current dataset shows gender distribution imbalance and represents primarily older adults (mean age approximately 65 years). The PD cohort also exhibits wide variation in disease duration (4.8 ± 4.0 years) and motor scores (MDS-UPDRS: 20.9 ± 7.9). These demographic imbalances and variabilities might affect the model's generalization ability, particularly when applied to underrepresented groups such as younger patients. Future studies should validate our approach on more demographically diverse populations, including younger patients and more balanced gender representation, to ensure broader applicability and reliability of the detection system.

From an implementation perspective, our method is well-suited for integration into existing wearable devices. The computational requirements are modest, with feature extraction and model inference capable of running on standard microprocessors found in commercial wearables. The sampling rate requirement of 31.25 Hz is readily achievable by most accelerometer sensors, and the single sensor approach significantly reduces hardware complexity and power consumption. For scalability, the system can be implemented in two ways: as a standalone device with local processing, or as a hybrid system where feature extraction occurs on-device and classification runs on a connected

smartphone. The latter approach enables easy model updates and data collection for continuous improvement while maintaining privacy through local processing of raw sensor data. Future development should focus on optimizing battery life for continuous operation, creating user-friendly interfaces for both patients and healthcare providers, and implementing secure data management protocols for long-term monitoring. Integration with existing health monitoring platforms and electronic health records would further enhance the system's clinical utility.

By pursuing the future directions outlined above, we can advance closer to realizing the full potential of wearable sensor technology for the early detection and continuous monitoring of Parkinson's disease. The ultimate goal is to contribute to improved patient care and enhanced quality of life for those affected by this condition.

V. CONCLUSION

Our study presents a novel ensemble learning approach for the detection of Parkinson's disease hand tremors during gait analysis based on data collected from a single wrist-worn sensor. This approach not only simplifies the detection process but also enhances the practicality and accessibility of monitoring for Parkinson's disease symptoms in real-time and in real-world settings. Our optimized ensemble learning model, which leverages new accelerometer-based features like Average Peak Interval Time (APIT) and Average Peak Frequency (APF), demonstrates superior performance with an accuracy of 86.67% and an AUC score of 95.56%. This indicates a significant improvement over existing methods, and it emphasizes the model's enhanced diagnostic accuracy and reliability. This approach could lead to more personalized and proactive management strategies, which could significantly impact PD treatment and patient quality of life.

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